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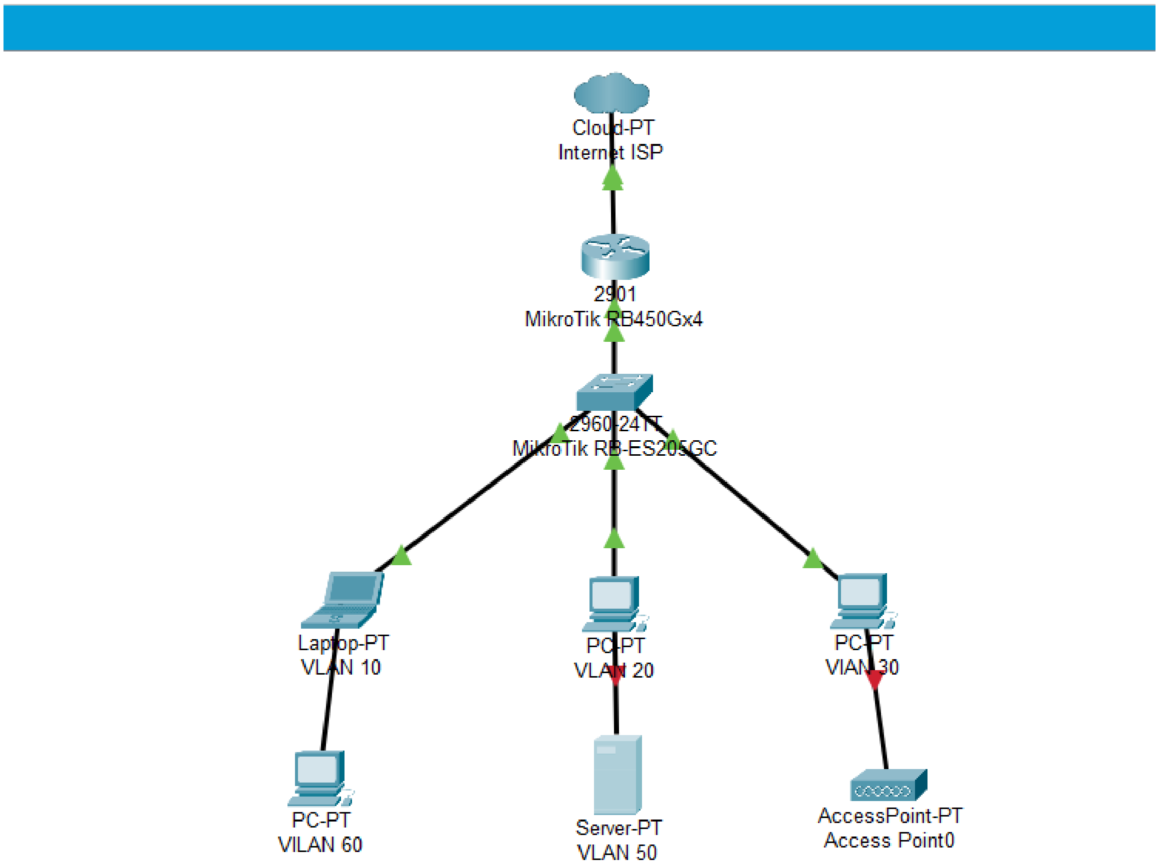
1.Introduction : In this project, I used Cisco Packet Tracer to practice how to set up VLANs and connect different parts of a network. I made several VLANs to separate each department or area so the network becomes more organized, secure, and easier to manage. My goal in this simulation was to create the VLANs, connect them correctly, and make sure all devices could still communicate with each other through inter-VLAN routing using the router-on-a-stick method.

2. Network Topology

The topology consists of one router (Router-Core), one main switch (Cisco 2960), and several PCs representing different VLANs. The logical structure is shown below:

Connection summary:
- Router-Core connected to Main-Switch via trunk link (G0/1)
- VLAN10: Yayasan (Room-1)
- VLAN20: Guru (Room-2)
- VLAN30: CCTV
- VLAN40: NVR

- VLAN50: Hotspot



3. VLAN and IP Address Plan

VLAN ID	VLAN Name	Gateway (Router Subinterface)	Example Device IP
10	Yayasan	192.168.10.1 /24	Room-1 = 192.168.10.10
20	Guru	192.168.20.1 /24	Room-2 =

			192.168.20.10
30	CCTV	192.168.30.1 /24	CCTV = 192.168.30.10
40	NVR	192.168.40.1 /24	NVR = 192.168.40.10
50	Hotspot	192.168.50.1 /24	Hotspot = 192.168.50.10

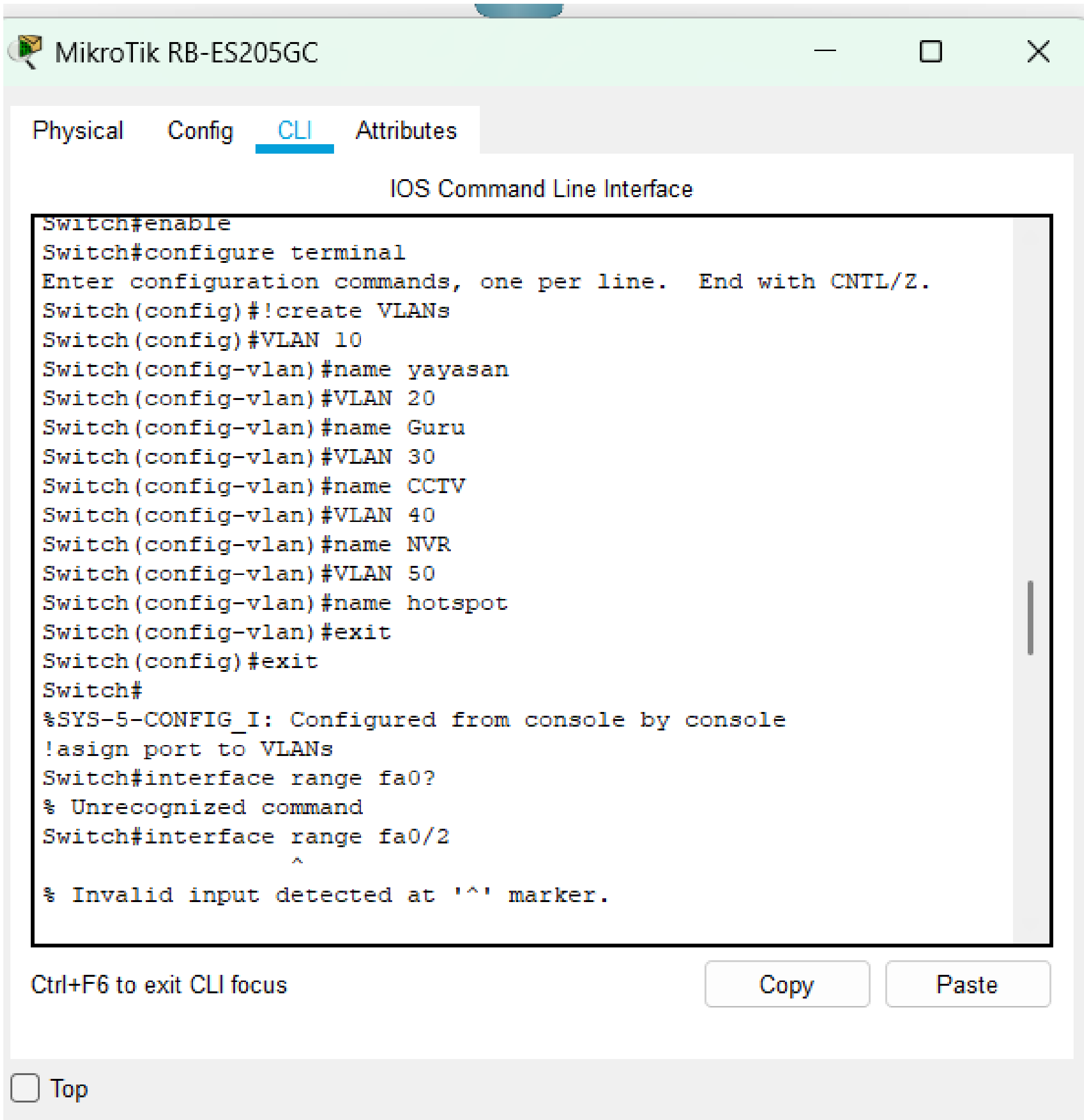
4. Switch Configuration

Below are the Cisco commands used to configure the switch. These commands create VLANs, assign ports, and configure the trunk port connected to the router.

```
enable
configure terminal
vlan 10
 name Yayasan
vlan 20
 name Guru
vlan 30
 name CCTV
vlan 40
 name NVR
vlan 50
 name Hotspot
exit

interface range fa0/2
 switchport mode access
 switchport access vlan 10
exit

interface range fa0/3
 switchport mode access
 switchport access vlan 20
exit

interface range fa0/4
 switchport mode access
 switchport access vlan 30
exit
```

```
interface range fa0/5
 switchport mode access
 switchport access vlan 40
exit

interface range fa0/6
 switchport mode access
 switchport access vlan 50
exit

interface gigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,50
 no shutdown
```

exit



5. Router Configuration (Router-Core)

The router uses subinterfaces on G0/1 to enable inter-VLAN routing.

enable
configure terminal

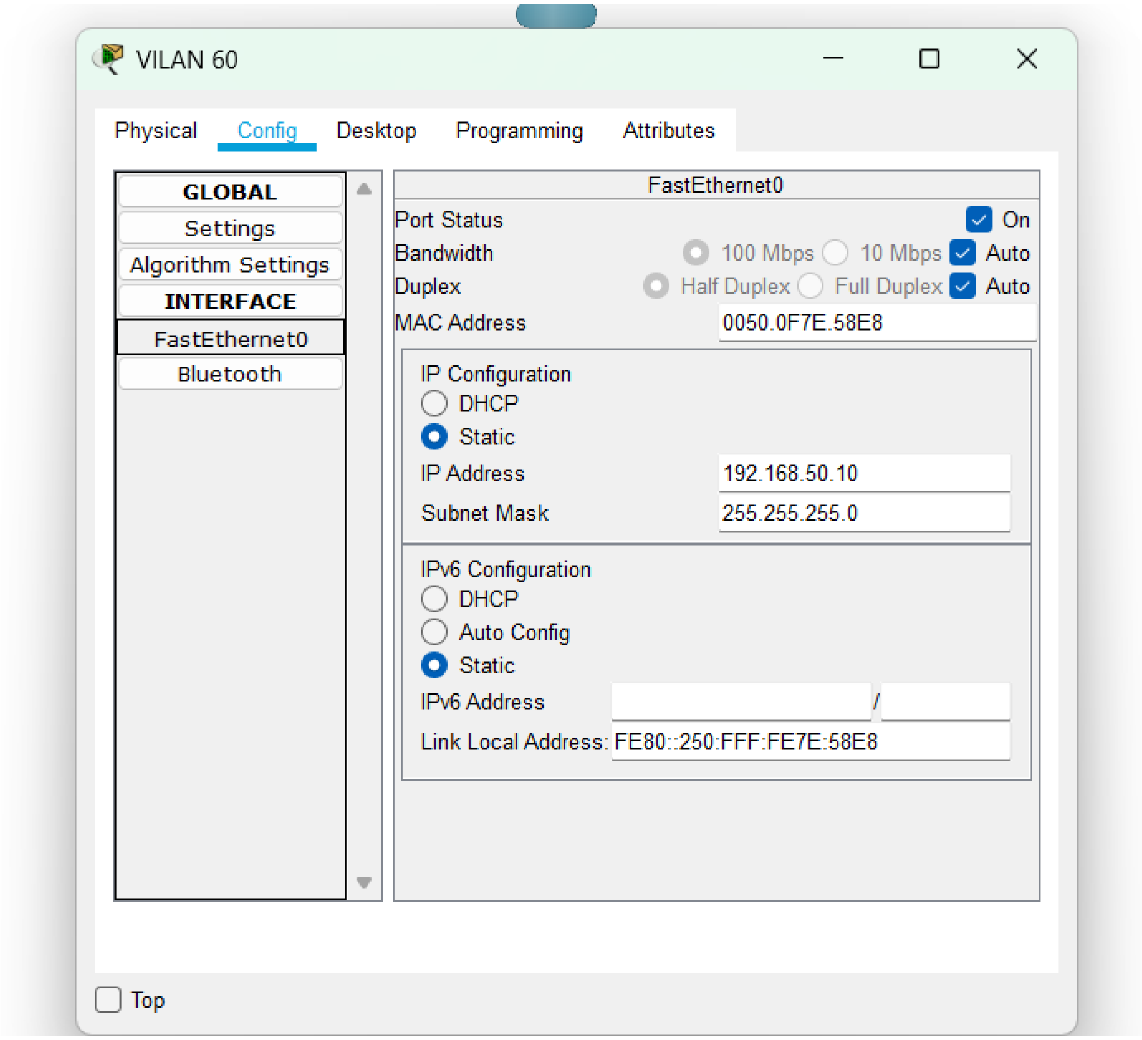
interface gigabitEthernet0/1
 no shutdown
 exit

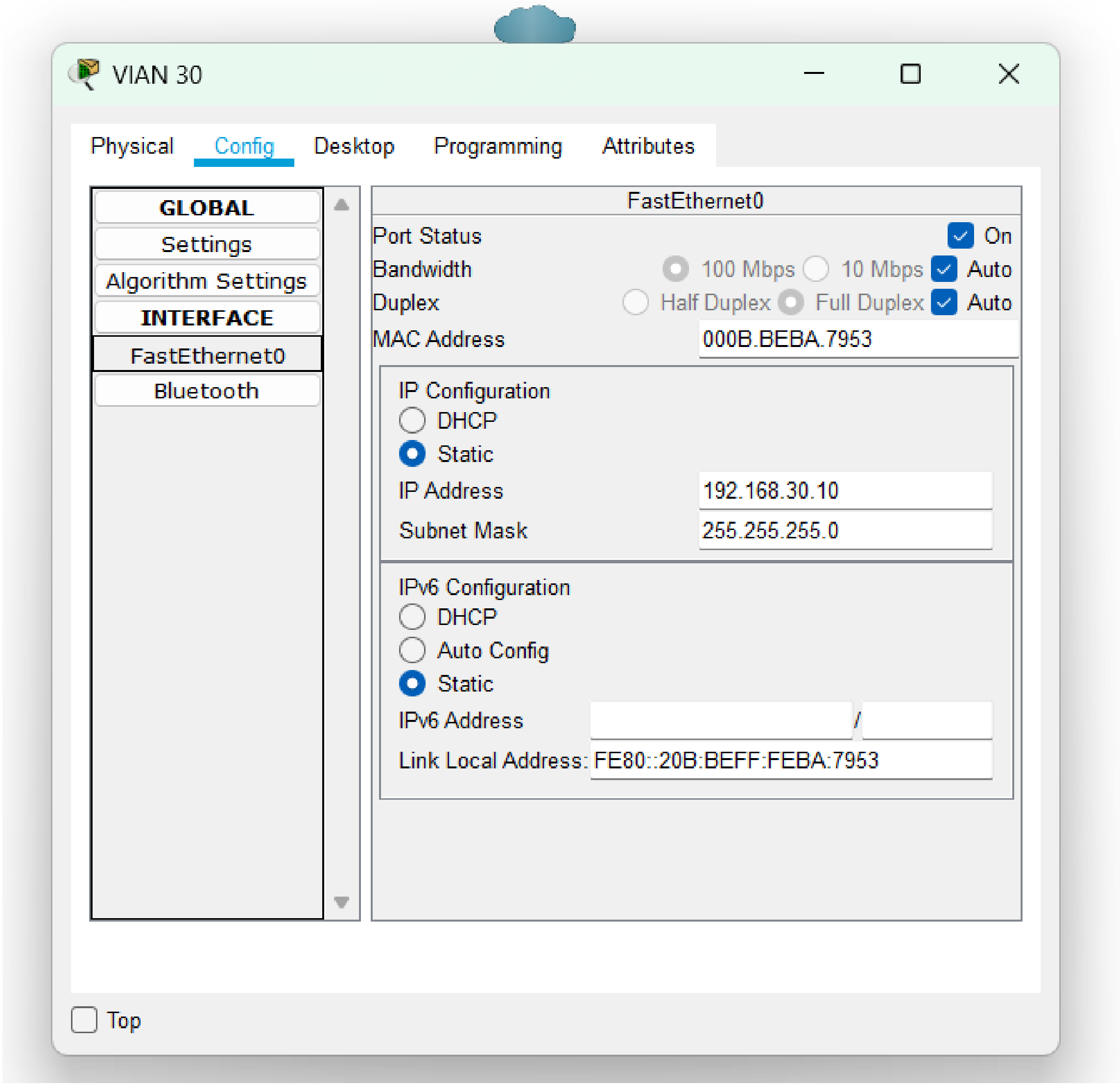
```
interface gigabitEthernet0/1.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit

interface gigabitEthernet0/1.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit

interface gigabitEthernet0/1.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit

interface gigabitEthernet0/1.40
 encapsulation dot1Q 40
 ip address 192.168.40.1 255.255.255.0
exit

interface gigabitEthernet0/1.50
 encapsulation dot1Q 50
 ip address 192.168.50.1 255.255.255.0
exit

end
write memory
```

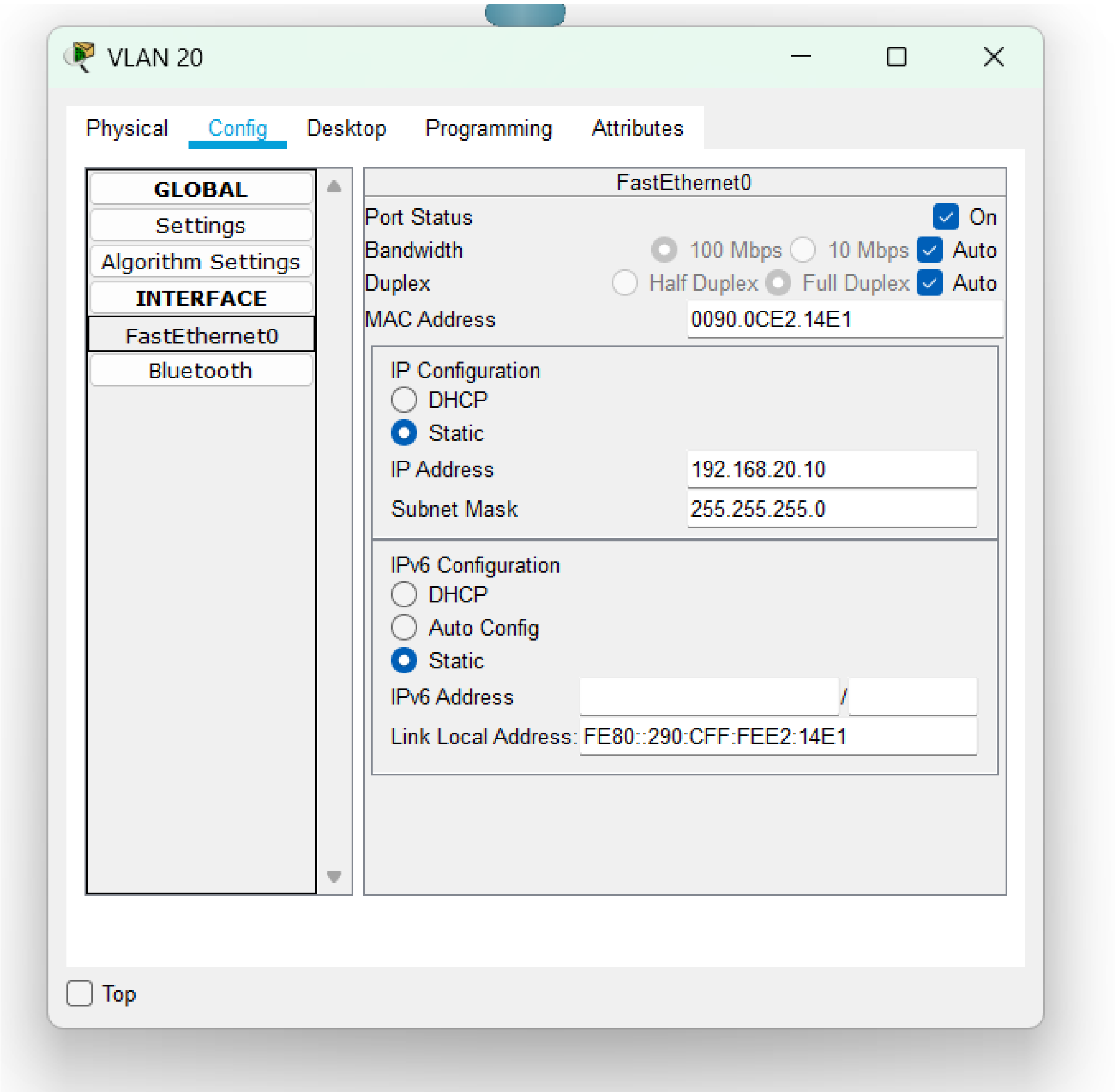
6. End Device Configuration

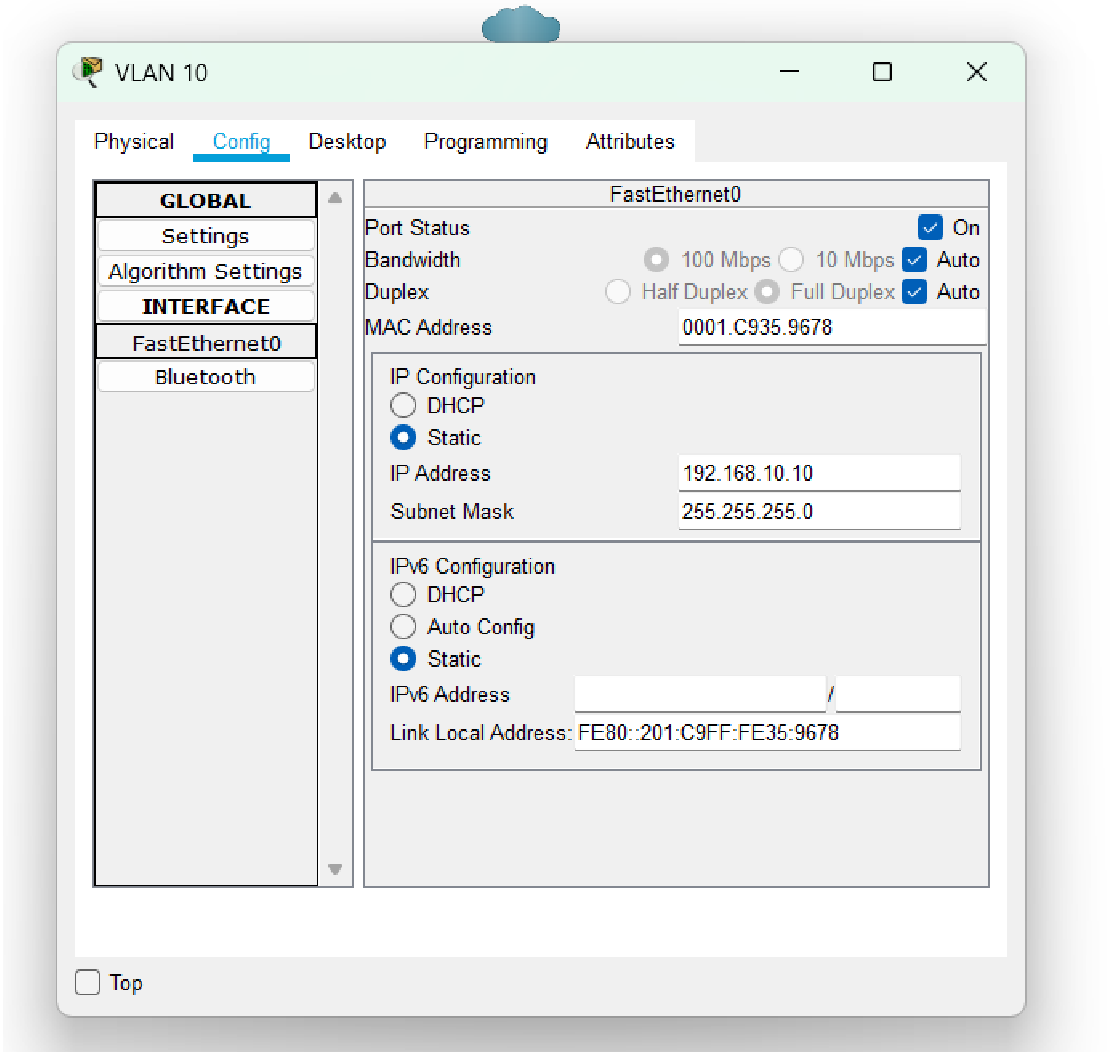
Each PC or end device is manually assigned an IP address that matches its VLAN network.
Example for Room-1:

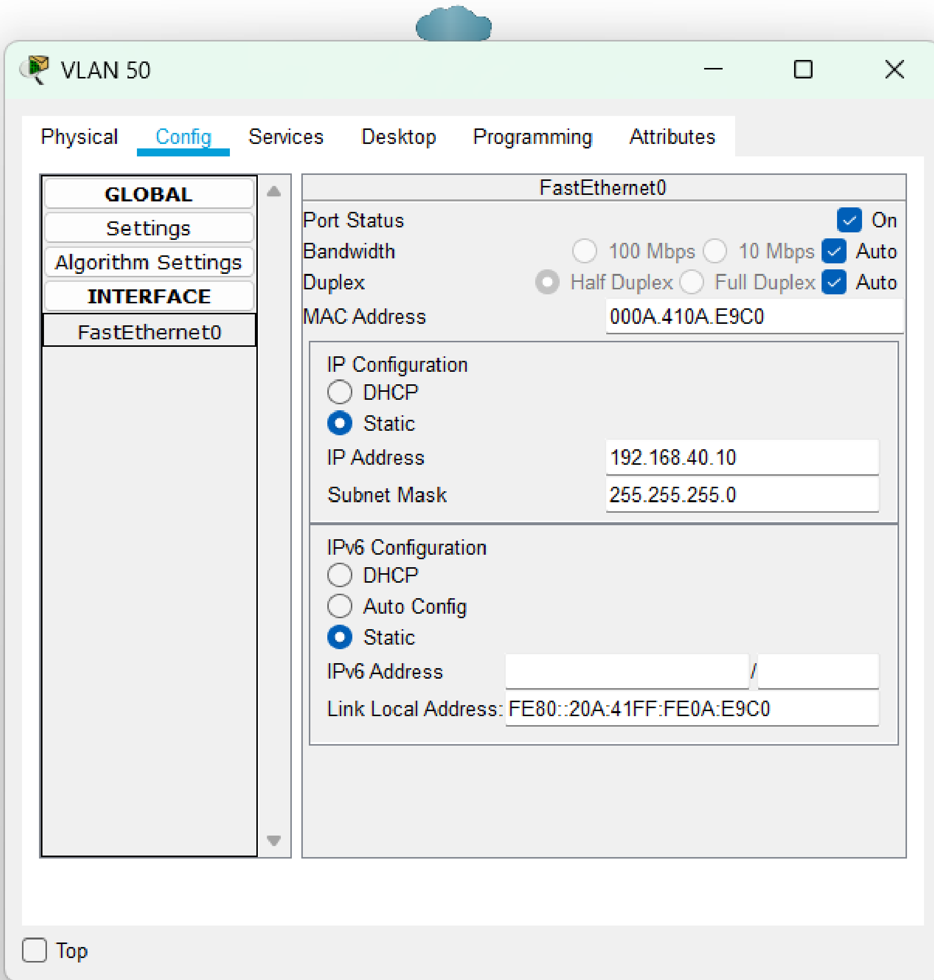
IP Address: 192.168.10.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1


VILAN 60
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0050.0F7E.58E8
IP Configuration
DHCP
Static
IP Address
192.168.50.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address:
FE80::250:FFF:FE7E:58E8
Top


VIAN 30
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
000B.BEBA.7953
IP Configuration
DHCP
Static
IP Address
192.168.30.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address:
FE80::20B:BEFF:FEBA:7953
Top


VLAN 20
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.0CE2.14E1
IP Configuration
DHCP
Static
IP Address
192.168.20.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address:
FE80::290:CFF:FEE2:14E1
Top


VLAN 10
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.C935.9678
IP Configuration
DHCP
Static
IP Address
192.168.10.10
Subnet Mask
255.255.255.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address:
FE80::201:C9FF:FE35:9678
Top



7. Conclusion

In this project, I learned how to create and manage VLANs using Cisco Packet Tracer. Each VLAN helped separate different parts of the network so that everything became more organized and secure. By using the Router-on-a-Stick method, I was able to make all VLANs communicate through one router interface.

This simulation really helped me understand how VLANs work in real situations and how important they are for network management. Overall, I can say that the configuration was successful because all the devices could connect and communicate properly between VLANs.